Simplify algebraic fractions

1 Simplify:

a $\dfrac{40y}{450}$ b $\dfrac{6n}{9n}$ c $\dfrac{42xy}{24y}$ d $\dfrac{15n^2}{-40n}$

e $\dfrac{64mn^2}{40m^2n}$ f $-\dfrac{10x^2}{6xy^3}$ g $\dfrac{6p^5}{3p^3}$ h $\dfrac{10a^3b^5}{25a^2b^9}$

2 Simplify:

a $\dfrac{25x-40}{55}$ b $\dfrac{4uv-20uw}{4u}$ c $\dfrac{5u}{5uv-30uw}$ d $\dfrac{9m^2+24mn}{18m^2}$

3 Simplify:

a $\dfrac{4x+20y}{x+5y}$ b $\dfrac{2x-6y}{x-3y}$ c $\dfrac{x-2}{2-x}$ d $\dfrac{5y-30}{6-y}$

e $\dfrac{2x+6}{x^2+3x}$ f $\dfrac{8-2x}{5x^2-20x}$ g $\dfrac{10x-15}{2x^3-3x^2}$ h $\dfrac{8-2xy}{x^2y-4x}$

i $\dfrac{12m^2+6mn}{18mn+36m^2}$

4 Simplify:

a $\dfrac{(x+3)(x-2)}{x+3}$ b $\dfrac{(2x-5)(x+4)}{x+4}$ c $\dfrac{x-7}{2(x-7)(x+4)}$ d $\dfrac{(x+6)(x+8)}{(x+6)(x-9)}$

e $\dfrac{x^2-8x+16}{x-4}$ f $\dfrac{x^2+7x+12}{x+4}$ g $\dfrac{x^2+9x+20}{(x+5)(x+1)}$ h $\dfrac{x^2-3x-18}{(x-6)(x+2)}$

5 Simplify:

a $\dfrac{m^2-16}{m-4}$ b $\dfrac{m^2-64}{8+m}$ c $\dfrac{x-10}{x^2-100}$ d $\dfrac{a^2-121}{11-a}$

6 Simplify:

a $\dfrac{8\,(p-q)^2}{8p^2-8q^2}$ b $\dfrac{x+10}{x^2+20x+100}$ c $\dfrac{x^2-4x+4}{7\,(x-2)}$ d $\dfrac{x^2+8x+15}{x+5}$

e $\dfrac{x+5}{x^2+2x-15}$ f $\dfrac{3x^2-18x-216}{4\,(x+6)}$ g $\dfrac{3y+6}{y^2-4}$ h $\dfrac{m^2-16}{5m+20}$

i $\dfrac{x^2+6x+9}{x^2-9}$ j $\dfrac{x^2-3x-10}{x^2-5x}$ k $\dfrac{x^2+5x+6}{x^2-2x-8}$ l $\dfrac{8\,(x+3)}{5x^2+30x+45}$

m $\dfrac{(x+3)\,(x-11)}{x^2-18x+77}$ n $\dfrac{4x^2+8x-192}{2x+16}$

7 Simplify:

a $\dfrac{xy + 10x + 9y + 90}{x + 9}$

b $\dfrac{5\left(x - 12\right)}{xy + 6x - 12y - 72}$

c $\dfrac{y - 8}{3xy - 24x - 30y + 240}$

d $\dfrac{5xy - 20x + 40y - 160}{8\left(y - 3\right)\left(y - 4\right)}$

e $\dfrac{wxy + wxz}{wy + wz + xy + xz}$

f $\dfrac{x^2 - xy + xz - yz}{x^2 - 2xy + y^2}$

8 Simplify the following algebraic fraction:

$$\frac{5\left(k^2 - 5\right)^4 + 20k\left(k^2 - 5\right)^5}{15\left(k^2 - 5\right)^4}$$